acceleration of gravity that they feel (which points from the one that ages the fastest to the one that ages the slowest). Then Einstein's law says that $g = Rc^2/D$. For the Pound-Rebca experiment in the Harvard tower, R was 210 picoseconds in one day, which is 2.43×10^{-15}, and the tower height D was 73 feet (22.3 meters). Inserting these into Einstein's law, we deduce $g = 9.8$ meters/second^2, which indeed is the gravitational acceleration on Earth.

Chapter 6. Gargantua's Anatomy

For a black hole such as Gargantua that spins extremely fast, the horizon's circumference C in the hole's equatorial plane is given by the formula $C = 2\pi GM/c^2 = 9.3\ (M/M_{\text{sun}})$ kilometers. Here M is the hole's mass, and $M_{\text{sun}} = 1.99 \times 10^{30}$ kilograms is the Sun's mass. For a very slowly spinning hole, the circumference is twice this size. The horizon's radius is defined to be this circumference divided by 2π: $R = GM/c^2 = 1.48 \times 10^8$ kilometers for Gargantua, which is very nearly the same as the radius of the Earth's orbit around the Sun.

The reasoning by which I deduce Gargantua's mass is this: The mass m of Miller's planet exerts an inward gravitational acceleration g on the planet's surface given by Newton's inverse square law: $g = Gm/r^2$, where r is the planet's radius. On the faces of the planet farthest from Gargantua and nearest it, Gargantua's tidal gravity exerts a stretching acceleration (difference of Gargantua's gravity between the planet's surface and its center a distance r away) given by $g_{\text{tidal}} = (2GM/R^3)r$. Here R is the radius of the planet's orbit around Gargantua, which is very nearly the same as the radius of Gargantua's horizon. The planet will be torn apart if this stretching acceleration on its surface exceeds the planet's own inward gravitational acceleration, so g_{tidal} must be less than g: $g_{\text{tidal}} < g$. Inserting the formulas above for g, g_{tidal} , and R,